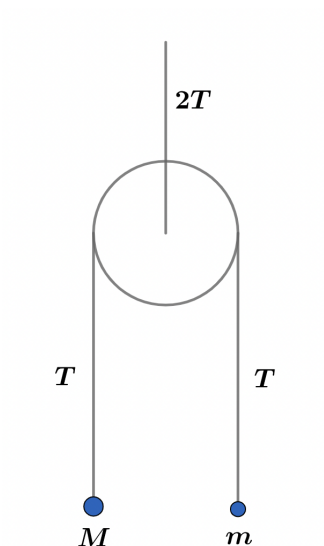


2022B F=ma Exam: Problem 7

Kevin S. Huang



Applying Newton's 2nd law to both masses, we have

$$Mg - T = Ma$$

$$T - mg = ma$$

Adding both equations,

$$(M - m)g = (M + m)a$$

$$a = \frac{M - m}{M + m}g$$

$$T = m(g + a) = m \left(1 + \frac{M - m}{M + m} \right) g = \frac{2Mmg}{M + m}$$

The scale reads

$$2T = \frac{4Mmg}{M + m} \leq (M + m)g$$

so the answer is A.